```
~ Create Database
CREATE DATABASE RACEDAY;
GO

~ Use correct Database to execute statements
USE RACEDAY;
GO

~ Create Tables Statements

~ 1. USER - Supertype for all registered RaceDay users (Event organisers & participants)
CREATE TABLE dbo.[USER]
(
    userID INT IDENTITY(1,1) PRIMARY KEY,
    firstName VARCHAR(50) NOT NULL,
    lastName VARCHAR(50) NOT NULL,
    email NVARCHAR(100) NOT NULL UNIQUE,
    password NVARCHAR(255) NOT NULL,
    phoneNumber NVARCHAR(20) NOT NULL,
    role VARCHAR(20) NOT NULL,
    profilePictureUrl NVARCHAR(500) NULL,
    registrationDate DATETIME2 NOT NULL DEFAULT GETDATE(),
    CONSTRAINT CK_USER_Role CHECK (role IN ('Organiser', 'Participant'))
);

~ 2. EVENT_ORGANISER - Subtype of USER, stores organisational details
CREATE TABLE dbo.EVENT_ORGANISER
(
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    event_organiserID INT PRIMARY KEY,
    organisationName NVARCHAR(100) NOT NULL,
    organisationContact NVARCHAR(100) NOT NULL,
    CONSTRAINT FK_EventOrganiser_User
    FOREIGN KEY (event_organiserID)
    REFERENCES dbo.[USER] (userID)
    ON DELETE CASCADE
);

-- 3. PARTICIPANT - Subtype of USER, stores participant profile details
CREATE TABLE dbo.PARTICIPANT
(
    participantID INT PRIMARY KEY,
    dateOfBirth DATE NOT NULL,
    gender NVARCHAR(20) NOT NULL,
    emergencyContactName NVARCHAR(100) NOT NULL,
    emergencyContactNumber NVARCHAR(20) NOT NULL,
    CONSTRAINT FK_Participant_User
    FOREIGN KEY (participantID)
    REFERENCES dbo.[USER](userID)
    ON DELETE CASCADE
);

-- 4. EVENT - Events created and managed by organisers
CREATE TABLE dbo.EVENT
(
    eventID INT IDENTITY(1,1) PRIMARY KEY,
    event_organiserID INT NOT NULL,
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    name NVARCHAR(150) NOT NULL,
    description NVARCHAR(500) NOT NULL,
    date DATE NOT NULL,
    location NVARCHAR(150) NOT NULL,
    distance DECIMAL(6,2) NOT NULL,
    startTime TIME NOT NULL,
    routemap NVARCHAR(255) NOT NULL,
    eventType NVARCHAR(50) NOT NULL,
    registrationDeadline DATETIME2 NOT NULL,
    bannerImageUrl NVARCHAR(500) NULL,
    CONSTRAINT FK_Event_Organiser
    FOREIGN KEY (event_organiserID)
    REFERENCES dbo.EVENT_ORGANISER(event_organiserID),
    CONSTRAINT CK_Event_Distance CHECK (distance > 0),
    CONSTRAINT CK_Event_Type CHECK (eventType IN ('Run', 'Walk', 'Cycle'))
);

-- 5. CATEGORY - Event categories, distance tiers, fees and capacity limits
CREATE TABLE dbo.CATEGORY
(
    categoryID INT IDENTITY(1,1) PRIMARY KEY,
    eventID INT NOT NULL,
    categoryName NVARCHAR(50) NOT NULL,
    distance DECIMAL(6,2) NOT NULL,
    categoryLevel NVARCHAR(50) NOT NULL,
    entryFee DECIMAL(10,2) NOT NULL,
    maxParticipants INT NOT NULL,
    CONSTRAINT FK_Category_Event
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    FOREIGN KEY (eventID)
    REFERENCES dbo.EVENT(eventID)
    ON DELETE CASCADE,
    CONSTRAINT UQ_Category_Event_Name UNIQUE (eventID, categoryName),
    CONSTRAINT UQ_Category_Category_Event UNIQUE (categoryID, eventID),
    CONSTRAINT CK_Category_Distance CHECK (distance > 0),
    CONSTRAINT CK_Category_EntryFee CHECK (entryFee >= 0),
    CONSTRAINT CK_Category_MaxParticipants CHECK (maxParticipants > 0)
);

-- 6. PARTICIPANT_EVENT - Associative entity resolving PARTICIPANT to EVENT many-to-many
CREATE TABLE dbo.PARTICIPANT_EVENT
(
    participant_eventID INT IDENTITY(1,1) PRIMARY KEY,
    eventID INT NOT NULL,
    categoryID INT NOT NULL,
    participantID INT NOT NULL,
    bib VARCHAR(20) NOT NULL,
    registrationDate DATETIME2 NOT NULL DEFAULT GETDATE(),
    registrationStatus NVARCHAR(30) NOT NULL DEFAULT 'Pending',
    CONSTRAINT FK_ParticipantEvent_Event
    FOREIGN KEY (eventID)
    REFERENCES dbo.EVENT(eventID),
    CONSTRAINT FK_ParticipantEvent_Category
    FOREIGN KEY (categoryID, eventID)
    REFERENCES dbo.CATEGORY(categoryID, eventID),
    CONSTRAINT FK_ParticipantEvent_Participant
    FOREIGN KEY (participantID)
```

```
    REFERENCES dbo.PARTICIPANT(participantID),
    CONSTRAINT UQ_ParticipantEvent_Registration UNIQUE (participantID, eventID),
    CONSTRAINT UQ_ParticipantEvent_Event_Bib UNIQUE (eventID, bib),
    CONSTRAINT CK_ParticipantEvent_Status
    CHECK (registrationStatus IN ('Pending', 'Confirmed', 'Cancelled', 'Completed'))
);

~ 7. RESULT - Official race results for participant enrolments
CREATE TABLE dbo.RESULT
(
    resultID INT IDENTITY(1,1) PRIMARY KEY,
    participant_eventID INT NOT NULL UNIQUE,
    time TIME NULL,
    position INT NULL,
    status NVARCHAR(30) NOT NULL,
    recordedDate DATETIME2 NOT NULL DEFAULT GETDATE(),
    CONSTRAINT FK_Result_ParticipantEvent
    FOREIGN KEY (participant_eventID)
    REFERENCES dbo.PARTICIPANT_EVENT(participant_eventID)
    ON DELETE CASCADE,
    CONSTRAINT CK_Result_Position CHECK (position IS NULL OR position > 0),
    CONSTRAINT CK_Result_Status CHECK (status IN ('Finished', 'DNF', 'DNS',
'Disqualified'))
);

~ 8. PERFORMANCE_HISTORY - Aggregated performance summary per participant
CREATE TABLE dbo.PERFORMANCE_HISTORY
(
    performance_historyID INT IDENTITY(1,1) PRIMARY KEY,
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```
    participantID INT NOT NULL UNIQUE,
    bestTimeRecord TIME NULL,
    totalWins INT NOT NULL DEFAULT 0,
    bestPositionHeld INT NULL,
    lastRace NVARCHAR(100) NULL,
    averagePosition DECIMAL(10,2) NULL,
    CONSTRAINT FK_PerformanceHistory_Participant
    FOREIGN KEY (participantID)
    REFERENCES dbo.PARTICIPANT(participantID)
    ON DELETE CASCADE,
    CONSTRAINT CK_PerformanceHistory_TotalWins CHECK (totalWins >= 0),
    CONSTRAINT CK_PerformanceHistory_BestPosition CHECK (bestPositionHeld IS NULL OR bestPositionHeld > 0),
    CONSTRAINT CK_PerformanceHistory_AveragePosition CHECK (averagePosition IS NULL OR averagePosition > 0)
);
GO


~ Insert Statements


~ 1. Users
INSERT INTO dbo.[USER] (firstName, lastName, email, password, phoneNumber, role, registrationDate)
VALUES
('Sipho', 'Khumalo', 'sipho.organiser@raceday.co.za', 'AQAAAAEAACcQAAAAEJ7u5bW4K9xL2pQ8mR3tY1vN6sH0fD5gC2bV4xZ8mQ==', '0821234567', 'Organiser', '2026-01-15 08:00:00'),
('Lerato', 'Mokoena', 'lerato.organiser@raceday.co.za', 'AQAAAAEAACcQAAAAED9fG2hJ5kL8zX1cV4bN7mQ0wE3rT6yU9iO2pA5sD8f==', '0839876543', 'Organiser', '2026-01-20 09:30:00'),
```

('John', 'Smith', 'john.participant@gmail.com',
'AQAAAAEAACcQAAAAEN3bV6cX9zL2pK5jH8gG1fD4sA7pW0eR3tY6u190P21K5==',
'0723456789', 'Participant', '2026-02-01 10:15:00'),

('Nomusa', 'Dlamini', 'nomusa.dlamini@gmail.com',
'AQAAAAEAACcQAAAAEI2wQ5rT8yU1oP4iL7kH0jG3fD6sA9pZ2xC5vB8nM1kL==',
'0765432109', 'Participant', '2026-02-05 14:20:00');

~ 2. Event Organisers (Fixed missing quote on email)

INSERT INTO dbo.EVENT_ORGANISER (event_organiserID, organisationName, organisationContact)

VALUES

(1, 'Gauteng Road Runners Club', 'info@gautengrunners.co.za'),

(2, 'Cape Endurance Events', 'contact@capeendurance.co.za');

~ 3. Participants

INSERT INTO dbo.PARTICIPANT (participantID, dateOfBirth, gender, emergencyContactName, emergencyContactNumber)

VALUES

(3, '1995-05-12', 'Male', 'Mary Smith', '0821112223'),

(4, '1998-08-22', 'Female', 'Bheki Dlamini', '0833334445');

~ 4. Events

INSERT INTO dbo.EVENT (event_organiserID, name, description, date, location, distance, startTime, routemap, eventType, registrationDeadline, bannerImageUrl)

VALUES

(1, 'Soweto Spring Marathon', 'Annual road running event through historic streets of Soweto.', '2026-08-15', 'Soweto, Johannesburg', 42.20, '06:00:00',
'https://racedaystorage.blob.core.windows.net/maps/soweto-map.png', 'Run', '2026-08-01 23:59:59', 'https://racedaystorage.blob.core.windows.net/banners/soweto-banner.jpg'),

(1, 'Joburg Fun Walk & Run', 'Family-friendly 5km and 10km event through Zoo Lake.', '2026-11-05', 'Zoo Lake, Johannesburg', 10.00, '07:30:00',

'https://racedaystorage.blob.core.windows.net/maps/zoolake-map.png', 'Walk', '2026-10-25 23:59:59', 'https://racedaystorage.blob.core.windows.net/banners/zoolake-banner.jpg'),

(2, 'Cape Peninsula Cycle Tour', 'Scenic cycling tour around the Cape Peninsula coastline.', '2026-12-01', 'Cape Town, Western Cape', 50.00, '05:30:00', 'https://racedaystorage.blob.core.windows.net/maps/cape-cycle-map.png', 'Cycle', '2026-11-15 23:59:59', 'https://racedaystorage.blob.core.windows.net/banners/cape-banner.jpg');

~ 5. Categories (Fixed missing quote on 'Intermediate')

INSERT INTO dbo.CATEGORY (eventID, categoryName, distance, categoryLevel, entryFee, maxParticipants)

VALUES

(1, 'Full Marathon 42.2km', 42.20, 'Advanced', 350.00, 1500),

(1, 'Half Marathon 21.1km', 21.10, 'Intermediate', 250.00, 3000),

(2, 'Zoo Lake 10km Challenge', 10.00, 'Open', 150.00, 500),

(3, 'Peninsula 50km Cycle', 50.00, 'Advanced', 450.00, 2000),

(2, 'Zoo Lake 5km Fun Walk', 5.00, 'Beginner', 100.00, 500);

~ 6. Participant Event Enrolments

INSERT INTO dbo.PARTICIPANT_EVENT (eventID, categoryID, participantID, bib, registrationDate, registrationStatus)

VALUES

(1, 2, 3, '1001', '2026-03-01 11:00:00', 'Completed'),

(2, 3, 3, '2005', '2026-03-02 12:30:00', 'Confirmed'),

(1, 1, 4, '1002', '2026-03-03 09:15:00', 'Completed');

~ 7. Results

INSERT INTO dbo.RESULT (participant_eventID, time, position, status, recordedDate)

VALUES

(1, '01:45:30', 12, 'Finished', '2026-08-15 11:30:00'),

(3, '03:58:45', 45, 'Finished', '2026-08-15 12:15:00');

-- 8. Performance History

```
INSERT INTO dbo.PERFORMANCE_HISTORY (participantID, bestTimeRecord, totalWins, bestPositionHeld, lastRace, averagePosition)
VALUES
(3, '01:45:30', 0, 12, 'Soweto Spring Marathon (Half Marathon 21.1km)', 12.00),
(4, '03:58:45', 0, 45, 'Soweto Spring Marathon (Full Marathon 42.2km)', 45.00);
GO
```